Acceleration means velocity is changing

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Acceleration / ES: aceleración / PT: aceleração

The rate velocity changes: how much velocity is gained or lost each second.

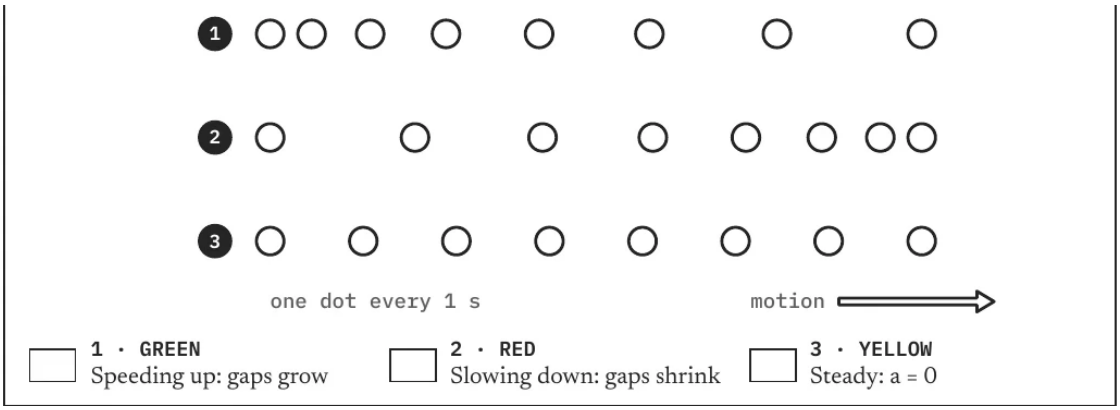
Also on your note page: Speeding up, Slowing down, Turning.

B READ THE RELATIONSHIP

Acceleration tells how quickly velocity changes.

Average acceleration = change in velocity ÷ time

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart's velocity changes from 0 to 8 m/s in 4 s. Find its acceleration.

Given $v_i = 0$ m/s $v_f = 8$ m/s $\Delta t = 4$ s

Rule $a = \Delta v \div \Delta t$

Work $\Delta v = 8 - 0 = 8$ m/s $a = 8 \div 4$

Answer 2 m/s² gains 2 m/s each second

E SAVE FOR THE FINAL MEETING

If XJ's velocity readings change from one night to the next, it is accelerating, and a forecast that assumes steady velocity will miss.

Read the physics: Acceleration

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Compare velocity before and after. A change in speed or a turn is a change in velocity. Divide the signed change in velocity by the time it took.

02 HOW IT WORKS

Read m/s² as a change in meters per second each second. Look at dot spacing at equal times: growing gaps mean faster motion; shrinking gaps mean slower motion.

03 WHAT TO CHECK

Compare velocity and acceleration. In a straight line, matching directions mean speed grows. Opposite directions mean speed falls until the object stops or reverses.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart's velocity changes from 0 to 8 m/s in 4 s. Find its acceleration. 2 m/s² gains 2 m/s each second.

05 CONNECT TO XJ

If XJ's velocity readings change from one night to the next, it is accelerating, and a forecast that assumes steady velocity will miss.

SAY IT BEFORE YOU WRITE

Can acceleration be negative while speed increases? The velocity changes from ___ to ___. Each second it changes by ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Acceleration

ES: aceleración / PT: aceleração

Meaning in my words	How can you tell?
How would you explain acceleration to a friend? In my words, this means ___.	A cart follows a bend at steady speed. Why is it accelerating? I can tell because ___.
Example or sketch	**Non-example and why**
Positive is right. A cart goes from −2 to −6 m/s in 2 s. What acceleration fits? My example shows ___.	A cart travels straight at −6 m/s the whole time. Does the minus sign mean it is accelerating? I would change ___ because ___.

Speeding up

ES: aumento de rapidez / PT: aumento de rapidez

Meaning in my words	How can you tell?
How would you explain speeding up to a friend? In my words, this means ___.	A cart changes from −2 to −5 m/s. What happens to its speed? I can tell because ___.
Example or sketch	**Non-example and why**
Draw three equal-time dots for a cart speeding up to the right. My example shows ___.	A cart moves right with acceleration to the left. Is it speeding up before it turns? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Slowing down

ES: disminución de rapidez / PT: redução de rapidez

Meaning in my words	How can you tell?
How would you explain slowing down to a friend? In my words, this means ___.	Velocity changes from −5 to −2 m/s. What happens to the speed? I can tell because ___.
Example or sketch	**Non-example and why**
Draw velocity and acceleration arrows for a bike braking while moving right. My example shows ___.	"Negative acceleration always means slowing down." What happens if velocity is negative too? I would change ___ because ___.

Turning

ES: cambio de dirección / PT: mudança de direção

Meaning in my words	How can you tell?
How would you explain turning to a friend? In my words, this means ___.	A rider keeps the speedometer at 4 m/s around a bend. Which part of velocity changes? I can tell because ___.
Example or sketch	**Non-example and why**
How would you draw two velocity arrows as a cart turns from east to north at steady speed? My example shows ___.	A cart brakes on a straight line and still moves east. Has it turned? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Labels you can use: before · after · change.

Before

After

Tell a partner

The velocity changes from ___ to ___. Each second it changes by ___.

On assignment: Acceleration

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two signed XJ velocities are −2 and −6 km/s, two seconds apart. The control room must choose its speed alert.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Telemetry engineer**

Send the average acceleration with its sign and unit. Sketch before and after velocity arrows so the shift lead can confirm the alert.

☐ **Replay commentator**

Narrate what changed over two seconds. Have your producer show which arrow proves whether XJ sped up or slowed down.

Word bank: before · after · change.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”